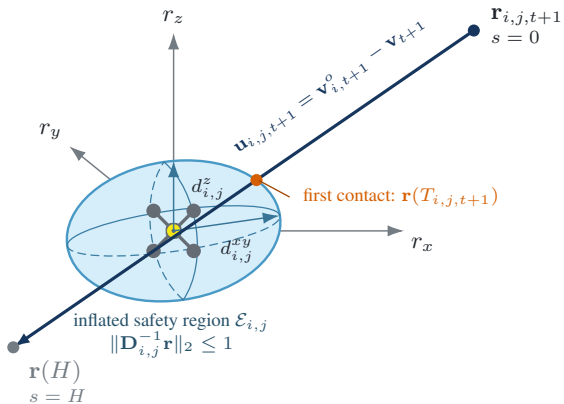


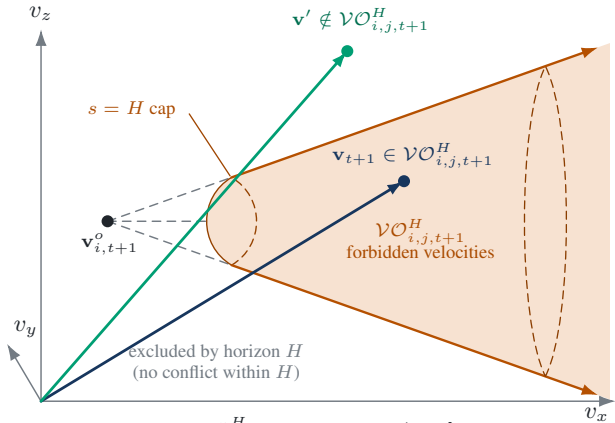
(a) Relative-space TTC test



$$\mathbf{r}(s) = \mathbf{r}_{i,j,t+1} + s\mathbf{u}_{i,j,t+1},$$

$$T_{i,j,t+1} = \min\{s \in (0, H] : \mathbf{r}(s) \in \mathcal{E}_{i,j}\}$$

(b) Finite-horizon 3D velocity obstacle



$$\mathbf{v} \in \mathcal{VO}_{i,j,t+1}^H \iff \exists s \in (0, H] :$$

$$\|\mathbf{D}_{i,j}^{-1} [\mathbf{r}_{i,j,t+1} + s(\mathbf{v}_{i,t+1}^o - \mathbf{v})]\|_2 \leq 1$$